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1.0 Introduction

The SC-300 system provides information to the operator through the use of display screens. Display screens represent the status of the process or machine being monitored/controlled by the Series Six programmable controller. Since this is the primary method of communications between the computer and the user, it is important to design an effective set of display screens for the application.

The SC-300 system supports from 1 to 96 different display screens. Each screen that is defined is given a number to identify it from all other screens in the system. Several screens are predefined by General Electric and are application independent. A list of these reserved screens appears later in this manual.

In order to configure screens, a set of utility programs have been provided as part of the SC-300 system. These programs allow drawing and defining the screens. This manual will describe the use of each of these programs to create display screens.

2.0 Screen Components

Each screen developed for an application is composed of three parts:

1.	Background data file	FOR??.IBM
2.	Foreground data file	DIS??.IBM
3.	Parameter linkage data file	SRN??.IBM

The question marks shown in the example filenames are used to indicate the screen number from 01 to 96.

Background Data File

This file consists of the textual information that never changes when the screen is displayed. Items like device nomenclature, simple character graphic diagrams, and data field titles would normally be considered background data.

Foreground Data File

This file defines each of the screen locations that will represent 'live' data. These 'fields' are refreshed at a user defined rate using information from the SC-300 database. A screen may contain from 0 to 128 'live' data fields.

Parameter Linkage Data File

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This file defines the tagname used for each 'live' data point in a screen. A tagname is a 1 to 9 character mnemonic that normally matches the nickname used in the Series Six program. For example, a 'live' data field on a screen is defined to represent the SC-300 database point EUR 0501. The Series Six stores this data into register 300 which is given a nickname of TMPFDBK. This file would provide the data required to 'link' TMPFDBK to EUR 0501. See appendix E for more information about the SC-300 database.

3.0 Creating Background Data Files

A utility program named SCDRAW.CMD has been provided to allow drawing the background information required by a display screen. This utility allows the user to create new background files or edit existing background files.

This program runs under the Concurrent DOS XM operating system (CDOS) in native mode. It is not possible to use this program under MSDOS or PCDOS.

This utility is invoked by typing its name (SCDRAW) at the CDOS prompt. A menu will appear on the top of the screen listing possible program operations. These commands include:

Dis - Loads a background data file into memory and displays it

Edit - Enters the drawing mode to allow editing of the screen data

Print- Provides the ability to print the screen contents to one of the several supported hard copy devices

Quit - Causes an exit back to the operating system. If a Save command was not performed, the program will ask if the screen should be saved prior to exiting.

Cfg - Allows configuration of the printer to be used for hard copy of developed screens.

Help - Displays a single screen of help information about all of the available commands.

Only the first character of the listed commands are significant. Extra characters following the first are ignored with the exception of the **D**isplay command. See details about the **D**isplay command for more information.

3.1 Display Command

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This command is used to load an existing background data file into memory prior to performing other program operations. If the requested file does not exist, the program will provide a warning message.

This command can be given with an optional parameter to eliminate the prompt that requests the screen number. The valid range for the screen number is from 1 to 96. Entries outside this range will provide a warning message.

Example: D <cr>
Background to display (1-96): 5 <cr>

OR

D5 <cr>

When this command is used, any previous background data resident in memory is lost and overwritten with new information.

3.2 Edit Command

This command enters a mode that allows drawing new screen background data. This mode is exited by pressing the **Escape** key. The function keys 1 to 10 are used to control various features of the drawing mode. These key operations are listed below:

F1 Change color to Black
F2 Change color to Blue
F3 Change color to Green
F4 Change color to Cyan
F5 Change color to Red
F6 Change color to Magenta
F7 Change color to Yellow
F8 Change color to White

F9 Select Foreground/Background for color changes
The selection of foreground or background for color changing is toggled between the two modes by function key F9. Initial startup mode is Foreground.

F10 Select Draw mode or graphics character set
The user can switch between Draw mode and the special characters in the IBM character set using the F10 toggle key. Draw mode is indicated by the word "Draw" in the upper left hand corner of the screen. Initial startup mode is "Draw".

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Several other keys are available to aid in drawing background displays. These include:

Home,Home Clear the screen to current background color
To clear the screen to build a new display, select the desired background color, then press the Home key twice. This will clear the entire user area of the screen.

End Drag the current color toggle
When in drag mode the current character under the cursor is re-colored to the current foreground/background color. Drag mode is toggled on and off with the End key and is indicated by a "D" under the Y coordinate when the drag mode is on.

PgUp, PgDn Scroll through IBM Character sets

These two keys will cause each of four character set displays to be drawn on the top of the screen when in the Character set mode (set by F10). Pressing the corresponding letter key (upper or lower case) will select either the top or bottom special character.

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Escape Exits Draw mode to main menu
When this key is pressed a prompt will appear asking if you want to Save or Rename the current display.

For example:

Save Changes (Y/N/Rename [Y])? Y <cr>

If the Rename option is chosen, a prompt will appear asking for the new background display number from 1 to 96.

For example:

Background to save (1-96) ? 5 <cr>

C A U T I O N

Because the save command looks at the screen memory directly, you must have the window that SCDRAW is running in set to the full-screen mode. That is, SCDRAW should take up the whole screen when the save command is being used. If SCDRAW does not cover the entire screen, press Ctrl-Del (the Ctrl and Del keys simultaneously) to expand the window back to full screen size before using the save command.

When the F10 key has been toggled and the program is in the "Draw" mode several development options are available to the user. The options will be listed at the top of the screen in place of the character set layout. Note that pressing the Shift key and then one of the function keys in unison adds ten to the number of the function key. In order to enter F11, for example, you would hold down the shift key and then press the F1 key at the same time. The following table lists the available drawing options:

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DRAWING OPTIONS

Key	Function
F11	IBM Thick Single Line Box See F13 description
F12	IBM Double Line Box. Makes a box using the current cursor position as the upper left hand origin. Use the cursor keys to size the box and then press enter or the space bar to leave the box mode.
F13	IBM Single Line Box. Makes a box using the current cursor position as the upper left hand origin. Use the cursor keys to size the box and then press enter or the space bar to leave the box mode.
F14	IBM Single Line. This function makes a line using the IBM character set. To make the line appear on the screen, use the cursor keys to draw. Press enter or the space bar to leave the line mode.
F15	ISC Line. (*Obsolete Command*)
F16	Oval. This function makes an oval that can be titled inside. Ovals can only be drawn horizontally and only the left and right cursor keys work with this drawing option. Press enter or the space bar to leave the oval mode.
F17	Define screen box. This function is used to outline an area of the screen using a box. The copy and clear box commands (described below) use this command to determine what screen coordinates to use. Same operation as the draw box command (F13).
F18	Copy Box. This command will copy the last box that was defined using the Define Box command (F17) to the current cursor position.
F19	Clear Box. This command will clear the marked box to the currently selected background color.
F20	Undo Command. Undo's last command.

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COLOR COMBINATIONS TO AVOID

BACKGROUND	FOREGROUND
RED	MAG becomes invisible CYN appears as white WHT appears as pink BLU clashes
GRN	CYN, WHT, or YEL becomes invisible MAG appears as RED
YEL	CYN, WHT, or GRN becomes invisible MAG appears as RED RED confused with MAG
BLU	YEL confused with WHT MAG confused with RED GRN confused with CYN, WHT WHT confused with YEL, CYN, GRN
MAG	RED becomes invisible YEL appears as ORG CYN confused with WHT GRN confused with CYN GRN clashes
CYN	YEL, GRN becomes invisible MAG confused with RED RED confused with MAG
WHT	CYN, YEL, GRN becomes invisible MAG confused with RED MAG, RED turn background pink

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3.3 Config Command

This command will present a numbered menu of printer options that can be selected similar to the list below:

```
Select Printer From The Following List

     0 = Epson Black & White  (default)
     1 = C.Itoh Color
     2 = IBM Color Ribbon
     3 = IBM Color Inkjet
     4 = IBM ProPrinter
     5 = HP Paintjet

Your Choice (0, 1, 2, 3, 4, or 5) ->
```

If a color printer is selected, an additional menu will be presented which allows you to adjust the technique used to print yellow colors. This menu is shown below:

```
Select Yellow Ribbon Substitution

     X = Do not substitute for yellow ribbon (default)
     M = Substitute Black for yellow ribbon
     N = Substitute Blue for yellow ribbon
     O = Substitute Red for yellow ribbon

Your choice (X, M, N, or O) ->
```

After completing the printer selection, an additional menu will be presented asking if all printer output should be directed to a file or the printer.

```
Select Output Destination Option

     F = Send output to file
     P = Send output to printer (default)

Your choice (F or P) ->
```

If the output to file option is chosen, the file used to store the printer image will be named FOR??.LIS where ?? is the background display number from 01 to 96.

After completing the configurations, all of your selections are stored in a file named SCPRINT.CFG. If configuration is not done, the program will automatically create this file and use the default settings shown above.

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3.4 Print Command

The print command can be used to obtain a hard copy record of the current display. Color printouts can be obtained if the required hard copy device is available.

3.4.1 Special Notes for HP PaintJet Color Printer

The printer driver software supplied for the HP PaintJet printer fully supports 16 color printouts with both foreground and background color representation. In addition, it can provide high quality color transparencies if desired.

When the HP PaintJet has been selected, the print command will allow printing any of the four virtual consoles instead of the current console. This allows the user to obtain printouts of other system programs (i.e. SC300 displays with function keys and alarm message).

A prompt will be given asking if paper or transparencies will be used. After selecting the appropriate media a prompt will be given asking for the virtual console number from 0 to 4. Answering 0 means use the current console. Entering a number from 1 to 4 will cause the program to switch the current virtual console to the new console and print the current screen contents.

If the virtual console is changed by this program, you must manually return to the original console after the hard copy printout has been completed.

The selection of yellow ribbon color substitution is not used when the HP PaintJet has been selected.

The HP PaintJet driver is only available in the SCDRAW utility.

C A U T I O N

Because the print command looks at the screen memory directly, you must have the window that SCDRAW is running in set to the full-screen mode. That is, SCDRAW should take up the whole screen when the print command is being used. If SCDRAW does not cover the entire screen, press Ctrl-Del (the Ctrl and Del keys simultaneously) to expand the window back to full screen size before using the print command.

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4.0 Creating Foreground Data Files

A utility program named SCLINK.COM has been provided to allow defining the 'live' data points for a display screen. This utility allows the user to create new 'live' data definitions or edit existing ones.

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Help - D

Dis - L
i

Edit - A

Seq - A

Link - A
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Print- Pr

Cfg -

Quit - Ex

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command for more

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4.1 Help Command

This command can be used to obtain a single page summary of all of the available commands in the SCLINK utility. Commands shown include Display, Edit, Sequence, Link, Print, Config, and Quit. Pressing any key will clear the help display and return back to the main program menu.

4.2 Display Command

This command will prompt for the 'Display to load' if a display number is not typed in the command line. Valid display numbers are from 1 to 96.

Example: D <cr>
Background to display (1-96): 5 <cr>

OR

D5 <cr>

When this command is used, any previous background or foreground data resident in memory is lost and overwritten with new information.

If the background and foreground files both exist, the command will load both files into memory and display the background file on the screen. The program will return to the main menu following this action whether the files were found or not. If a display is loaded, the upper right corner of the screen will show the current display number.

Zeroes will be shown at every 'live' data location using the configured display representation (symbol).

It is possible to view the characteristics of each configured 'live' data point by using the **Tab** and **Shift-Tab** key combinations. Pressing **Tab** will advance you to the next configured point while **Shift-Tab** while backtrack one point.

4.3 Edit Command

This command causes a new menu to appear showing all of the available editing commands. Each of these commands are listed and described below:

4.3.1 Help Command

This command can be used to obtain a single page summary of all of the available commands in the Edit mode. Commands shown include Add, Delete, Change, Move, and

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Exit. Pressing any key will clear the help display and return back to the edit menu.

4.3.2 Exit Command

This command will cause an exit back to the main menu of the program. If any changes were made during the editing session, a prompt will be given asking if you want to save your changes. Respond with either a 'Y' or 'N' and press return. If no changes were made then no prompt will be given.

4.3.3 Change Command

This command allows the user to change any of the characteristics of a specific 'live' data point. Each of these characteristics are displayed on the bottom of the screen. Pressing return leaves the current characteristic unchanged.

Name: This is the nickname that should be used to derive the actual SC-300 database location. This nickname can be from 1 to 9 alpha-numeric characters in length. Failure to assign a nickname will prevent the parameter linkage process (described later) from working. Valid nicknames may include letters and numbers. No special characters should be used. Blank characters are not allowed.

Symbol: This is a numeric code that designates how to display the 'live' data. Appendix A in this document lists all of the available symbol codes and their display respresentations.

Color: This characteristic defines the foreground and background colors to use for the 'live' data field. Using the up/down cursor keys will show the various foreground colors while the left/right keys will show background colors. There are four special colors indicated by the numbers 128-131. These color codes are used when the 'live' data is to change colors for an alarm condition.

Code	Description
128	Digital point on blue background
129	Digital point on black background
130	Analogs on blue background
131	Analogs on black background

To define the actual colors used for each of

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these four codes, use the SETCOLOR utility. See Appendix C in this manual for further information on color codes.

Change: This characteristic defines whether the user will be able to change or modify this 'live' point from this screen. Acceptable answers are 'Y' or 'N'.

Security: This characteristic defines the security level required to change this point if the changeable status is 'Y'. Acceptable answers range from '0' to '9'. If the current user is logged on with a point level security set at 3, then only points with security levels 0, 1, 2, and 3 can be modified by this user. Please note that this 'live' data characteristic applies only to this screen. Other screens may define different security levels for exactly the same point. Security levels of zero cause the point to have NO security checks performed.

Suffix: This characteristic is used in conjunction with certain display symbol codes. Acceptable responses include any three digit integer. See Appendix A for further description of suffix usage.

After all changes have been made, a prompt is given asking if all of your entries are complete for this point. If not, answer 'N' and you will be allowed to change the point again.

4.3.4 Move Command

This command allows you to re-position any 'live' data point to another set of XY coordinates. When selecting this command, you use the cursor keys to move to the new desired location and press return. If you should decide to change your mind you can abort this command by pressing the Escape key.

Before selecting this command be sure to position the cursor on the point that is to be moved.

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4.3.5 Add Command

This command is used to define new 'live' data points on the screen. It provides a series of prompts designed to obtain all of the necessary characteristics for the point.

Upon requesting this command the program will ask you to move the cursor to the screen location where the 'live' data is to be displayed. This should be the origin point for the desired symbol representation.
The rest of the prompts are exactly the same as those used in the **Change** command. When you have completed all of your data entries, you will be asked if you are finished. If so, the program will return to the edit menu.

The new 'live' data point will be shown on the screen as a question mark at the XY coordinates that you chose.

4.3.6 Delete Command

This command allows you to delete 'live' data points that have already been configured for the screen. Prior to using this command you should position the cursor over the question mark associated with this point. Cursor positioning is done using the **Tab** and **Shift-Tab** key combinations. After selecting the Delete command from the menu, the system will ask for confirmation. If you enter 'Y' and press return the point will be deleted.

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4.4 Sequence Command

This command allows adjustment of the visitation order used by the **Tab** and **Shift-Tab** key combination to move the cursor. This visitation order is used by both this program and the on-line SC-300 program. Typical usage of this command is to optimize the cursor positioning to reduce the number of tabs to change frequently accessed data.

To specify the visition order you must select each visitation point in the order you wish to visit them. The Select command is provided to perform this selection. Using the Tab key, position the cursor to each point and choose the Select menu option. The top of the screen will show how many parameters ('live' points) have been selected out of the total that are defined. Continue selecting points until all of them have been selected. Use the Quit command to save the new sequence. If not all of the points have been sequenced, the program will NOT save your sequence changes.

If a mistake is made, the Deselect command can be used to remove the least recently selected point from the list of sequenced points.

4.5 Link Command

This command is provided to link the generated foreground data file with the project memory map file (MMP). See description of the Parameter Linkage File in this manual for further details. See section 5.0.

4.6 Quit Command

This command will exit the program back to the operating system or menu from which it was invoked. The program does not ask to save anything even if changes were made to the foreground data definitions.

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5.0 Creating Parameter Linkage Files

A utility program named SCLINK.COM has been provided to allow defining the 'live' data points for a display screen. This utility allows the user to create new 'live' data definitions or edit existing ones.

This program runs under the Concurrent DOS XM operating system (CDOS) in PCDOS emulation mode. It is not recommended to use this program under MSDOS or PCDOS as the program has not been tested in these environments. Prior to running this utility the following command **MUST** be given or incorrect program operation may result:

```
D>8087=ON
D>SCLINK
```

This utility is invoked as shown above. A menu will appear on the top of the screen listing possible program operations.

When creating a foreground data file using the SCLINK utility, two files were actually being created. The first file contains all of the 'live' point characteristics excluding the nickname for the point. This is the DIS??.IBM file. The second file created contains the nickname of each live data point along with its sequence number. This is the SRN??.IBM file.

In order for the SC-300 program to actually display 'live' data from the database, it is necessary to link the 'live' data to the project memory map.

The project memory map is contained in the ProjectName.MMP file (downloaded from the factory VAX software development system). This memory map file contains entries that associate a nickname to the Series Six register location and the SC-300 database location. Whenever this memory map changes, the foreground data files must be re-linked to this file.

5.1 Link Command

The **Link** command shown on the main menu of the SCLINK utility program performs the work required to adjust the foreground data file so that each 'live' data point points to the correct SC-300 database location.

Errors and warnings are shown on the screen during this process. A serious error would occur when a nickname used for a screen point is not defined in the project memory map. Another error would occur when no SRN file exists for the screen.

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1.0 Appendix A - Display Symbol Codes

The SC-300 system supports representing 'live' data in a variety of ways. Each of these display representations can be specified using a **symbol** code number. This appendix describes each of the available symbol codes.

I M P O R T A N T

For some applications of the SC-300 software not all of these symbol codes may be available.

Symbol codes from 00 to 99 are used to indicate a numeric representation. The first digit of this two digit code indicates the number of digits to display in front of the decimal point. The second digit indicates the number of digits to show after the decimal point. When the first digit is greater than 4, no digits following the decimal will be displayed.

Hence, a symbol code of 32 will display a number in the format 999.99 and a symbol code of 41 will display a number in the format 9999.9. Integer values should be specified using a zero in the second digit. A decimal point will not be shown when a zero is specified as the second digit.

Keep in mind that only six digits can be represented. In other words, an entry of 66 is not legal; whereas an entry of 33 or 24 is acceptable.

An exception to the six digit constraint is the symbol code 80. This can be used to show large real numbers with no decimal points. In particular, this symbol is often used to show double precision data such as long-term production data.

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2.0 Text, Graphic, and Special Symbols

Code Description

200 Horizontal valve with actuator and status indication of OPEN or CLOSED. This symbol is normally not provided with the standard SC-300 system.

201 Digital ON or OFF indication. This symbol uses three characters and fills the ON string with a leading blank.

202 Trend Display

203 Trend Display which does not have the time on the X-Axis.

204 Trend Display which does not have the X or Y axis labeled. This is normally used when using a trend to indicate the level in a tank.

205 Horizontal bar which is one character high and 20 characters wide. The value displayed must be pre-scaled to a span of 0 to 100. The bar indicates 0 on the left and 100 on the right. This symbol is normally not provided with the standard SC-300 system.

207 Hex display. This symbol can be used to display an integer value as a four digit hex number followed by the letter H. For example, an integer value of 4 would look like this: '0004 H'.

208 This symbol is reserved for use by the Drive Control Screen.

210 This symbol is reserved for use by the Drive Control Screen.

211 Pump symbol with an ON or OFF status. The status is shown as three characters above the pump. This symbol is normally not provided with the standard SC-300 system.

212 Horizontal valve with status indication of OPEN or CLOS. This differs from symbol 200 because it does not have an actuator on top. This symbol is normally not provided with the standard SC-300 system.

213 Simple vertical valve with a status indication of OPEN or CLOSE. This symbol is normally not provided with the standard SC-300 system.

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Code Description

215 Displays integer values as two character ASCII. The least significant byte of the integer is displayed first as the leftmost character. For example, if the integer value were 3132H, this symbol would display '21' on the screen.

216 If the value is 1.0, this symbol displays a single character 'H'. If the value is not equal to 1.0, a 'C' is displayed. This symbol normally is used to indicate Heating and Cooling for plastics applications.

217 This symbol draws a vertical bar. It has been designed to be grouped together to form a vertical bar graph. The value to be graphed should be in the range from 0 to 20.0. Values outside this range are clamped. The height of the bar is 10 character cells.

218 Displays 'SEL' if the value equals 1.0 or else three blank characters.

219 Displays the alternate tab symbol. This symbol is either '=> ' or three blank characters. The symbol only shows when the screen point is currently tabbed (highlited). This symbol is normally not used for application screens.

221 Displays 'ALM' if the value equals 1.0 or else three blank characters.

222 Displays 'ENA' if the value equals 1.0 or else three blank characters.

223 Displays circle for use in making indicating lights. The color of the light changes when the point goes into and out of alarm conditions. (Assumes color codes 128-131.)

224 Displays a single ASCII character if value is equal to 1.0. If the value is not equal to 1.0, a blank is displayed. The single ASCII character to be displayed is obtained from the display symbol suffix defined for the point.

225 A universal horizontal bar graph. This symbol relies on the Series Six to preformat the value into an encoded value. The low byte specifies the bar length. The high byte specifies the bar offset from the left hand XY origin. The display suffix defines the total bar length (maximum X value minus minimum X value). Bar measurements are in half units (0-160). Useful for making GANTT chart displays or machine timing displays.

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Code Description

226 Displays 'MANUAL' if the value equals 1.0 or else 'AUTO '.

227 Displays a horizontal bar 8 character cells long. The color of this bar is red when the input point is set. When the input is zero, the color changes to green. The colors used by this symbol are not user-definable.

228 This symbol displays a vertical bar graph using a ten character cell area. The origin is at the bottom of the bar. The input value should be in the range of 0.0 to 20.0 in order to get the maximum resolution of the bar graph.

229 Similar

271 Displays the tagname of the point. Use of this symbol requires the system to have this point configured as a memory resident tagname. See System Configuration User's manual for more information on creating memory resident tagnames. (This symbol is not available in SC300 versions later than Aug. 26, 1988).

272 Displays the tag description of the point. Use of this symbol requires the system to have this point configured as a memory resident tagname. See System Configuration User's manual for more information on creating memory resident tagnames. (This symbol is not available in SC300 versions later than Aug. 26, 1988).

273 Displays the tag units for analogs or set/reset mnemonics for digital points. If the digital point is set, the set mnemonic will be shown, otherwise the reset mnemonic will be used. Use of this symbol requires the system to have this point configured as a memory resident tagname. See System Configuration User's manual for more information on creating memory resident tagnames.

274 Displays the tagname, description, value, and either units for analogs or set/reset mnemonics for digitals. Use of this symbol requires the system to have this point configured as a memory resident tagname. See System Configuration User's manual for more information on creating memory resident tagnames. (This symbol is not available in SC300 versions later than Aug. 26, 1988).

277 Displays free form analog outputs. This symbol will attempt to show as much significant data as can fit into a 6 character field. The number of decimal places depends on the number of digits required in front of the decimal point.

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1.0 Appendix B - Reserved Display Screens

Several of the display screens are pre-defined by General Electric and should not be modified except as indicated below. The following paragraphs list each of these screens and what modifications, if any, may be made by the user.

Display **Description**

1 Initial program startup display. This screen can be modified to reflect the proper customer data. It is always used as the initial startup display. In addition, the F10 key will cause this display to be shown.

16 Input/Output Status Display. This screen can be used to display a user selected input table and output table.

17 Input/Output/Register Diagnostic Display. This screen can be used to display a snapshot of user defined inputs, outputs, and registers from the Series Six.

60 - 69 Super Drive Control Screens. These screens are reserved for use by the Drive display system. They are created automatically using the MAKESMD utility program and should never be modified. The actual number in use depends on the number of drives in the system and the specific number defined per screen. Please note that only one background data file is used by all of these screens --- FOR60.IBM. The SCLINK program should not be used to manipulate these screens.

88 This screen is reserved for use by the recipe system. It is used to display recipe names.

89 This screen is reserved for use by the recipe system. It is used for the recipe compare/edit system.

90 This screen is reserved for use by the recorder system. It is used to show the current pen connections and their setup(s).

95 System Status Display. This display shows the current status of many of the programs that run in background. This screen is useful for monitoring the SC-300 system operation.

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1.0 Appendix C - Color Tables

C O L O R C O M B I N A T I O N S

0 - 63	Constant colors
64 - 127	Blinking colors
128	Digital on blue background
129	Digital on black background
130	Analogs on blue background
131	Analogs on black background

CODE	BACK	FORE	CODE	BACK	FORE
0	black	black	32	blue	black
1	"	red	33	"	red
2	"	green	34	"	green
3	"	yellow	35	"	yellow
4	"	blue	36	"	blue
5	"	magenta	37	"	magenta
6	"	cyan	38	"	cyan
7	"	white	39	"	white
8	red	black	40	magenta	black
9	"	red	41	"	red
10	"	green	42	"	green
11	"	yellow	43	"	yellow
12	"	blue	44	"	blue
13	"	magenta	45	"	magenta
14	"	cyan	46	"	cyan
15	"	white	47	"	white
16	green	black	48	cyan	black
17	"	red	49	"	red
18	"	green	50	"	green
19	"	yellow	51	"	yellow
20	"	blue	52	"	blue
21	"	magenta	53	"	magenta
22	"	cyan	54	"	cyan
23	"	white	55	"	white
24	yellow	black	56	white	black
25	"	red	57	"	red
26	"	green	58	"	green
27	"	yellow	59	"	yellow
28	"	blue	60	"	blue
29	"	magenta	61	"	magenta
30	"	cyan	62	"	cyan
31	"	white	63	"	white

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DECIMAL	OCTAL	HEX	ASCII		
0	000	00	NUL	null	(CTRL-@)
1	001	01	SOH	start of heading	(CTRL-A)
2	002	02	STX	start of text	(CTRL-B)
3	003	03	ETX	end of text	(CTRL-C)
4	004	04	EOT	end of transmission	(CTRL-D)
5	005	05	ENQ	enquiry	(CTRL-E)
6	006	06	ACK	acknowledge	(CTRL-F)
7	007	07	BEL	bell	(CTRL-G)
8	010	08	BS	backspace	(CTRL-H)
9	011	09	HT	horizontal tab	(CTRL-I)
10	012	0A	LF	linefeed	(CTRL-J)
11	013	0B	VT	vertical tab	(CTRL-K)
12	014	0C	FF	form feed	(CTRL-L)
13	015	0D	CR	carriage return	(CTRL-M)
14	016	0E	SO	shift out	(CTRL-N)
15	017	0F	SI	shift in	(CTRL-O)
16	020	10	DLE	data link escape	(CTRL-P)
17	021	11	DC1	device control 1	(CTRL-Q)
18	022	12	DC2	device control 2	(CTRL-R)
19	023	13	DC3	device control 3	(CTRL-S)
20	024	14	DC4	device control 4	(CTRL-T)
21	025	15	NAK	negative acknowledge	(CTRL-U)
22	026	16	SYN	synchronous idle	(CTRL-V)
23	027	17	ETB	end of transmission	(CTRL-W)
24	030	18	CAN	cancel	(CTRL-X)
25	031	19	EM	end of medium	(CTRL-Y)
26	032	1A	SUB	substitute	(CTRL-Z)
27	033	1B	ESC	escape	(CTRL-[)
28	034	1C	FS	file separator	(CTRL-\)
29	035	1D	GS	group separator	(CTRL-])
30	036	1E	RS	record separator	(CTRL-^)
31	037	1F	US	unit seperator	(CTRL-_)
32	040	20	SP	space	
33	041	21	!		
34	042	22	"		
35	043	23	#		
36	044	24	$		
37	045	25	%		
38	046	26	&		
39	047	27	'		
40	050	28	(		
41	051	29	)		
42	052	2A	*		
43	053	2B	+		
44	054	2C	,		
45	055	2D	-		
46	056	2E	.		
47	057	2F	/		
48	060	30	0		

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DECIMAL	OCTAL	HEX	ASCII	DECIMAL	OCTAL	HEX	ASCII	
49	061	31	1	97	141	61	a	
50	062	32	2	98	142	62	b	
51	063	33	3	99	143	63	c	
52	064	34	4	100	144	64	d	
53	065	35	5	101	145	65	e	
54	066	36	6	102	146	66	f	
55	067	37	7	103	147	67	g	
56	070	38	8	104	150	68	h	
57	071	39	9	105	151	69	i	
58	072	3A	:	106	152	6A	j	
59	073	3B	;	107	153	6B	k	
60	074	3C	<	108	154	6C	l	
61	075	3D	=	109	155	6D	m	
62	076	3E	>	110	156	6E	n	
63	077	3F	?	111	157	6F	o	
64	100	40	@	112	160	70	p	
65	101	41	A	113	161	71	q	
66	102	42	B	114	162	72	r	
67	103	43	C	115	163	73	s	
68	104	44	D	116	164	74	t	
69	105	45	E	117	165	75	u	
70	106	46	F	118	166	76	v	
71	107	47	G	119	167	77	w	
72	110	48	H	120	170	78	x	
73	111	49	I	121	171	79	y	
74	112	4A	J	122	172	7A	z	
75	113	4B	K	123	173	7B	<	
76	114	4C	L	124	174	7C	|	
77	115	4D	M	125	175	7D	>	
78	116	4E	N	126	176	7E	~	
79	117	4F	O	127	177	7F	DEL	delete
80	120	50	P					
81	121	51	Q					
82	122	52	R					
83	123	53	S					
84	124	54	T					
85	125	55	U					
86	126	56	V					
87	127	57	W					
88	130	58	X					
89	131	59	Y					
90	132	5A	Z					
91	133	5B	[					
92	134	5C	\					
93	135	5D	]					
94	136	5E	^					
95	137	5F						
96	140	60	`					

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Appendix E - SC-300 Database Description

The SC-300 system maintains a memory resident database of real-time data. This database reserves specific memory for each type of data required by the system. The following table lists each type available, the name used to refer to the type, and its storage requirements in bytes.

Type	Name	Bytes	Range
Real	EUR	8	10^{-307} to 10^{+307}
Integer	EUI	2	+32767 to -32768
Digital Inputs	DII	1	0 or 1
Digital Outputs	DOO	1	0 or 1
Raw Count Inputs	SCA	2	+32767 to -32768
Raw Count Outputs	OUT	2	+32767 to -32768
Flags	FLG	1	0 or 1

The standard SC-300 system supports a total of 2048 discrete entries for each type listed above. The first 512 of each type are reserved for use by the SC-300 run-time system while the remaining 1536 are available for use in applications.

Real and integer values are scaled from the raw values obtained from the Series Six into actual engineering units. Raw values are integer values which have not been scaled in any way from the values received from the Series Six.

Alarms can only be configured for the following datatypes:

EUR
EUI
DII
DOO

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